Project Part 1: Experimental Design and Planned Hypothesis Testing

Graduate students only (BIOL 5860)

This is the first of four checkpoint deliverables for the graduate project (see the Assignments overview). Use your own research data if at all possible; data from the published literature is also acceptable if you don't yet have your own. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas by the end of Week 7. See the syllabus for the late policy.

1. Research question and system

Describe the biological system and the question you're investigating. What gap or uncertainty does this address?

2. Hypotheses

State your null and alternative hypotheses in plain language, and, where applicable, in terms of a testable statistical prediction (e.g., "mean X differs between groups," "Y increases with X").

3. Design

Using the seven-question framework from Weeks 5–6:

1. What is the experimental (or observational) unit?
2. What are the treatments/predictors, and how many levels does each have?
3. How many units per treatment (replication) — actual or planned?
4. Is/was assignment to treatment randomized? If this is an observational study, say so explicitly.
5. Is there blocking, matching, or other control for a nuisance variable?
6. What is the response variable, and how is/was it measured?
7. If you're using an existing dataset, does its analysis (as published) match its design?

4. Planned analysis

Based on the design above and what you've learned so far this semester, which statistical test or model do you plan to use, and why? It's fine if this changes later — this is a plan, not a commitment.